

Mathematics A

Unit 1 • Chapter OL-T54 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

4 classified items • 8 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T54 • Expertise Q16+ • 4 items • 8 marks

Chapter	Items	Marks	Starts
01. OL-T54 Histograms	4	8	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Histograms

Unit 1 • Expertise • 8 marks • 4 item(s)

June 2022 | 4MA1-1H Q21 | 3 marks

Estimate a partial-class frequency

June 2021 | 4MA1-1H Q18 | 2 marks

Complete a table from a partial histogram

June 2021 | 4MA1-1H Q18 | 1 marks

Complete missing histogram bars from mixed information

June 2021 | 4MA1-1H Q18 | 2 marks

Estimate a histogram proportion, fraction or percentage

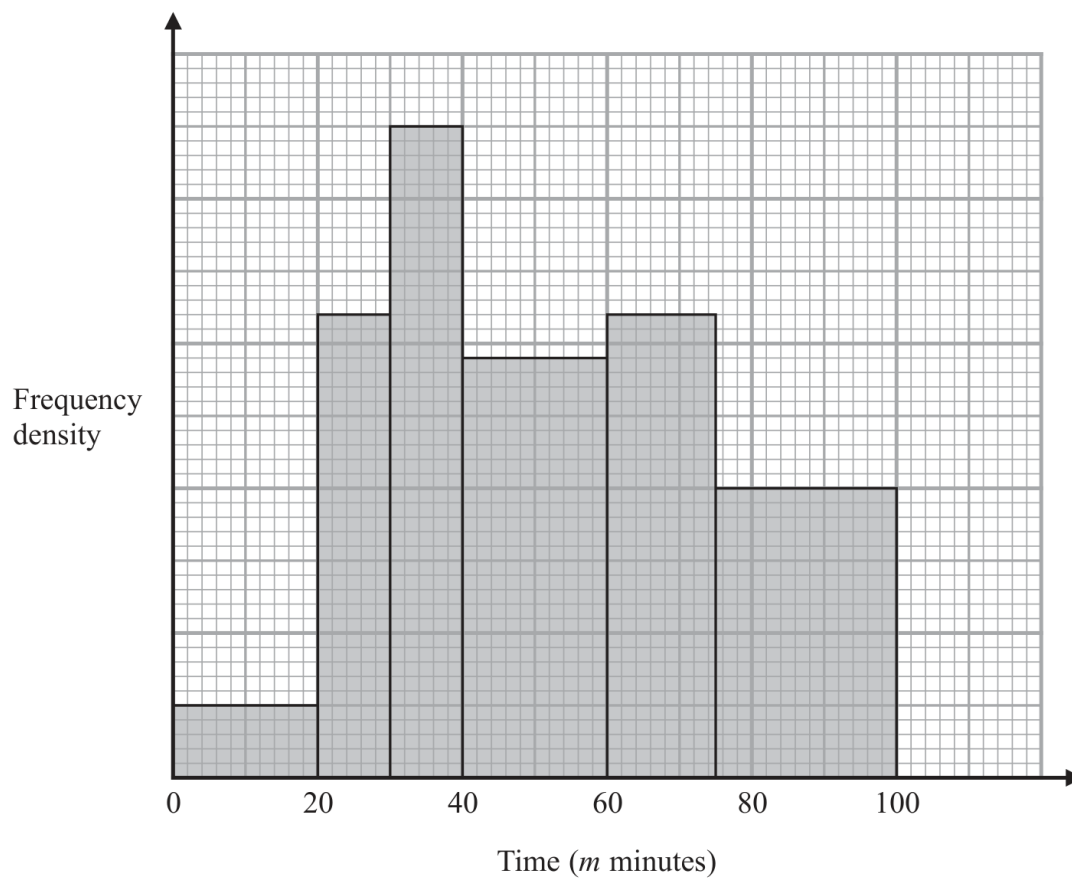
Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-1H Q21 Q21

3 marks

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

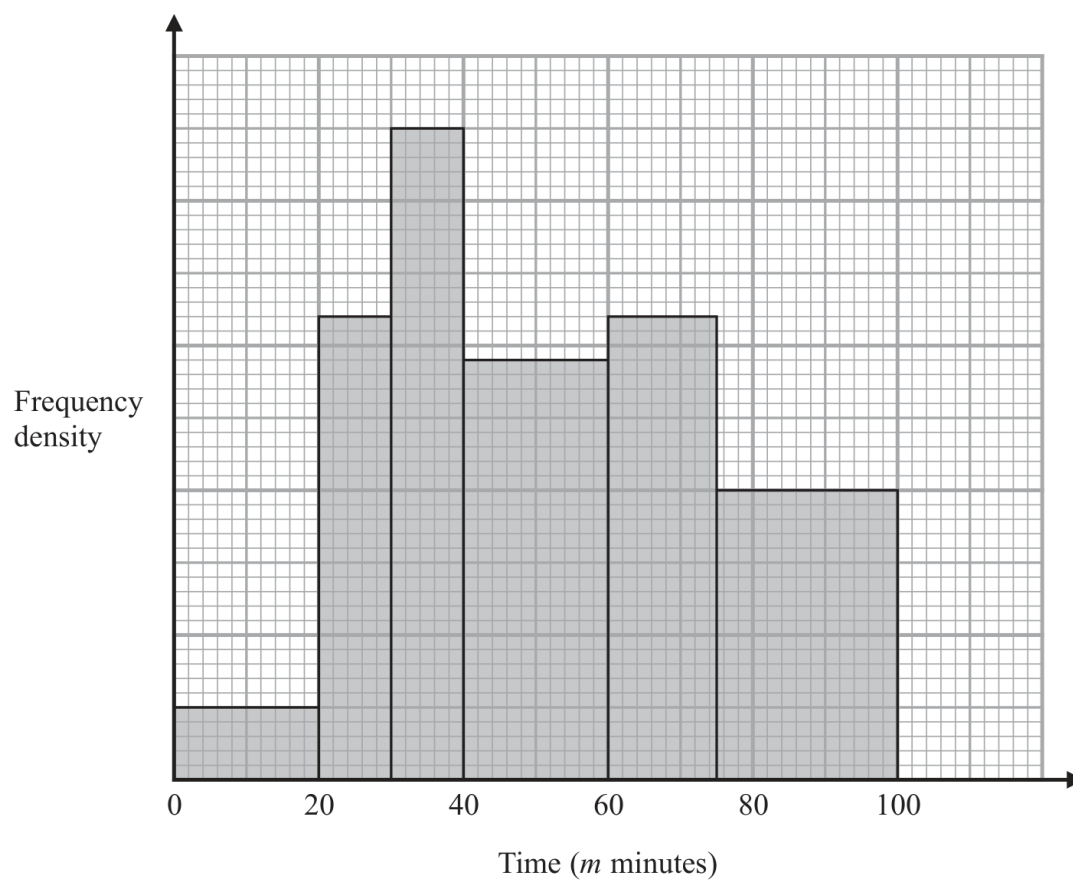
There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

4MA1-1H Q21 Q21

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Chapter: Histograms

3 marks

Q21 – Chapter: Histograms

Family: Interpret histogram areas • Variant: Estimate a partial-class frequency

Method

For $0 < m \leq 20$, frequency density = $10/20 = 0.5$; use this bar to calibrate the vertical scale.

Why: Histogram area equals frequency, and density equals frequency divided by class width.

Method

From the calibrated grid, the densities over 50-60, 60-75, and 75-80 are approximately 2.9, 3.2, and 2.0.

Why: Split the requested interval wherever a histogram class boundary occurs.

Method

Estimated frequency = $10(2.9) + 15(3.2) + 5(2.0) = 29 + 48 + 10 = 87$ children.

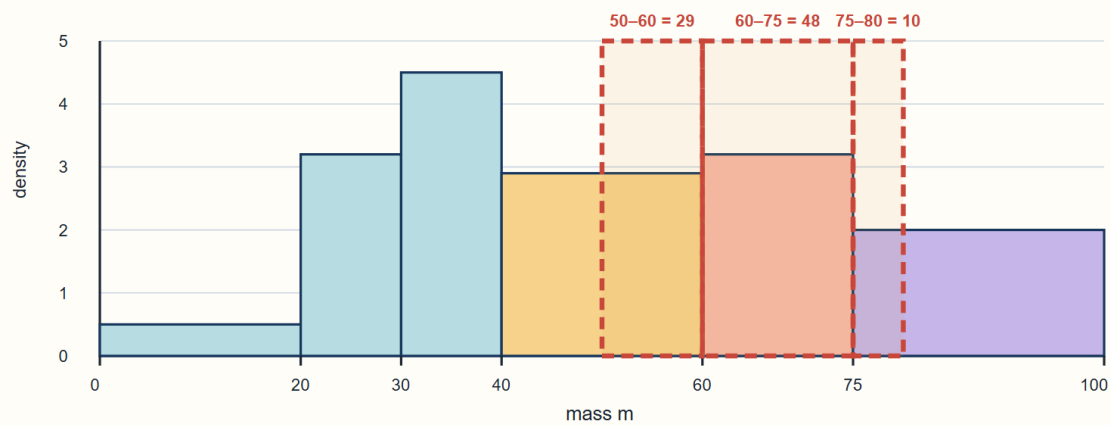
Why: Add the areas of the three relevant histogram pieces.

Final answer

estimated children: 87

Chapter: Histograms

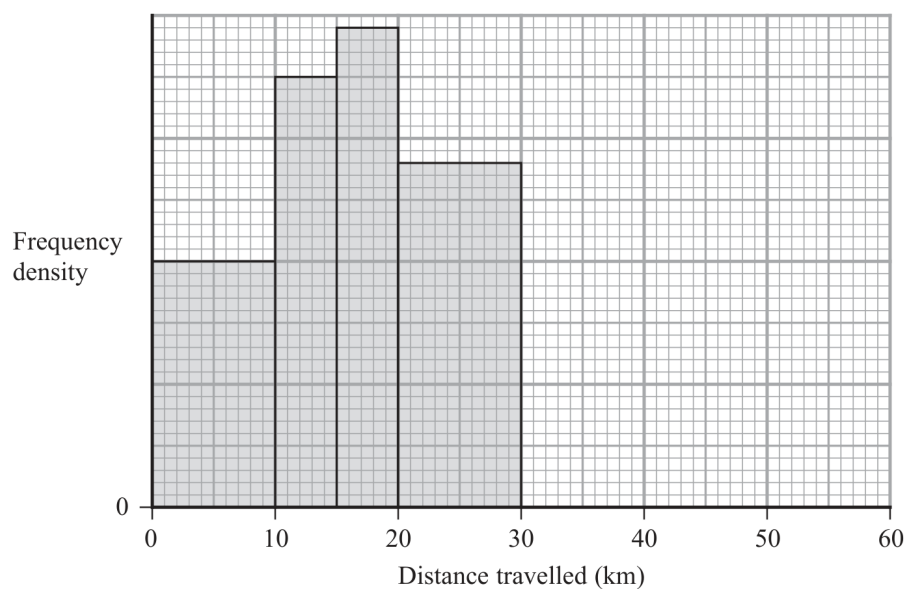
3 marks

Q21(a) Annotated histogram readingDensity $\times$ class width gives frequency; split 50–80 at 60 and 75Target estimate: $10 \times 2.9 + 15 \times 3.2 + 5 \times 2.0 = 87$ Check: frequency = density $\times$ width

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- 18 The table and histogram give information about the distance travelled, in order to get to work, by each person working in a large store.

Distance (d km)	Frequency
$0 \leq d < 10$	40
$10 \leq d < 15$	
$15 \leq d < 20$	
$20 \leq d < 30$	
$30 \leq d < 60$	30



Using the information in the table and in the histogram,

- (a) complete the table,

(2)

- (b) complete the histogram.

(1)

One of the people working in the store is chosen at random.

- (c) Work out an estimate for the probability that the distance travelled by this person, in order to get to work, was greater than 25 km.

(2)

Worked solution

June 2021 | Question 18 | 2 marks

Family: Construct and complete histograms • Variant: Complete a table from a partial histogram

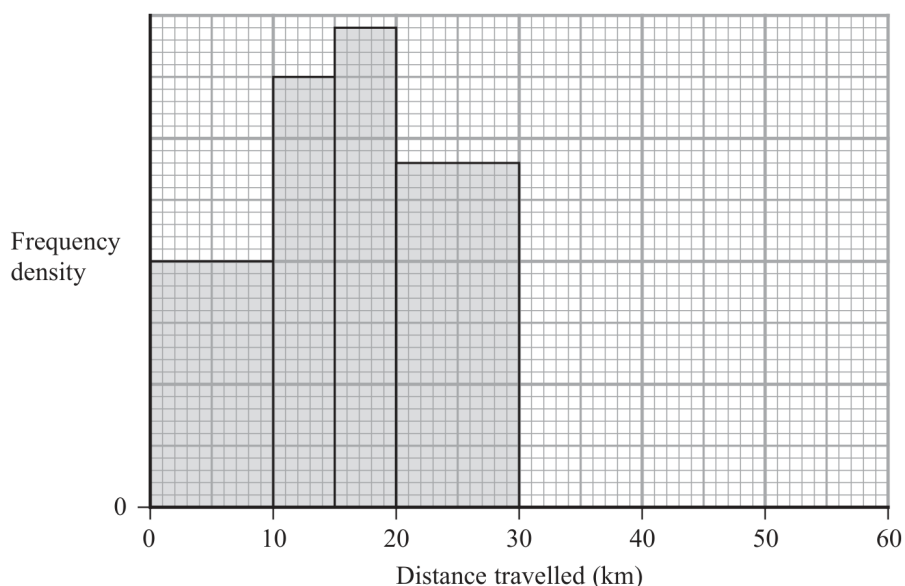
Method / working

1. Use frequency density = frequency / class width to complete the missing table values 35 and 39.
2. Read the remaining frequency from the histogram scale: 56.

Final answer: 35, 39, 56

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Worked solution

June 2021 | Question 18 | 1 marks

Family: Construct and complete histograms • Variant: Complete missing histogram bars from mixed information

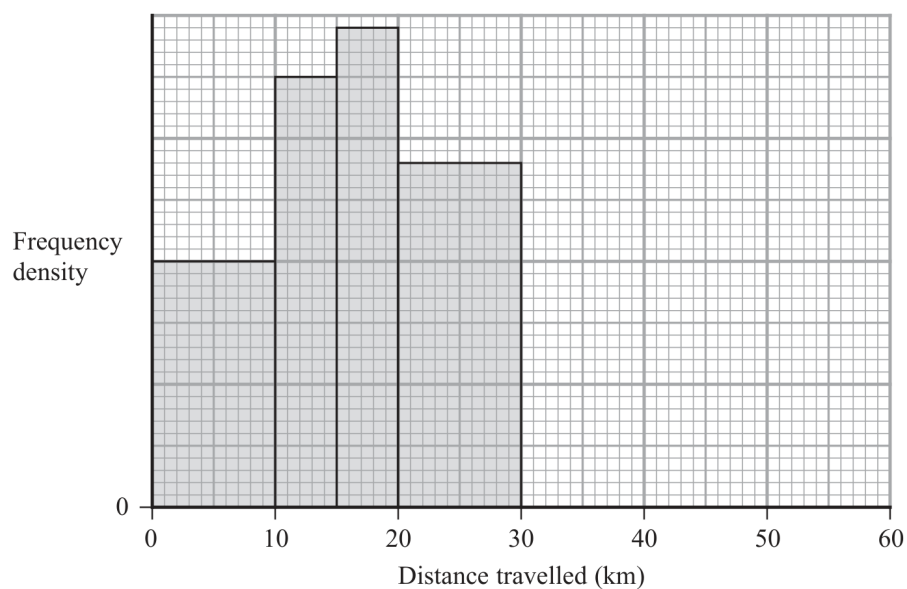
Method / working

1. Complete the 30-60 class with the bar height 1 on the frequency-density scale.

Final answer: Height 1 for $30 \leq d < 60$

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Worked solution

June 2021 | Question 18 | 2 marks

Family: Interpret histogram areas • Variant: Estimate a histogram proportion, fraction or percentage

Method / working

1. Estimated count above 25 km = $0.5 \times 56 + 30 = 58$ (the 30-60 class contributes 30; do not add the already-used table frequencies again).

2. Total count = 200, so probability = $58/200 = 0.29$.

Final answer: 0.29